

Rate, Leakage, and Distortion for Gaussian Sources with an Informed Encoder

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Abstract

An encoder observes a jointly Gaussian vector and describes part of it to a decoder that sees the description alone, under a weighted quadratic distortion. A second party holds a noisy linear observation of the source and never decodes. Two costs attach to one description: its rate, and its conditional leakage to the second party, the information the description reveals beyond what that party already observes, which is also the least rate at which that party can re-encode the description. The region of jointly achievable rate and leakage is known in single-letter form from the secure source coding literature, and the minimum leakage of a vector Gaussian source is known when the rate is unconstrained, but the joint Gaussian region has not been evaluated. We evaluate it when the decoder has no side information, for every distortion weight and every observation map of the second party. Both costs are convex functions of the error covariance of the description, so the whole rate–leakage frontier is a convex program with unique minimizers, and the minimum leakage is a determinant maximization. When the optimal description is one-dimensional, certified by an explicit test, the minimum leakage is a closed-form function of the largest generalized eigenvalue of a matrix pencil in which the distortion budget enters, and its eigenvector is the direction the description reads. That direction turns as the budget changes unless it is shared by the geometries of the decoder and the second party, which no description built for a single party exhibits. The rate-optimal and leakage-optimal descriptions differ whenever the second party’s observation touches the decoder’s principal direction. For independent components the leakage-optimal split of the distortion is not reverse water-filling and can describe different components from the rate-optimal split. For a scalar described variable the pencil returns an explicit closed form.

Index Terms

Equivocation, Gaussian sources, information leakage, rate–distortion theory, secure source coding, side information.

I. INTRODUCTION

One description of data is often held by more than one party. The decoder that reconstructs from it pays its rate. A second party that holds the description together with its own observation of the source learns from it the conditional mutual information between the source and the description given that observation. In a secrecy reading the second party is an eavesdropper and this quantity is the information leakage beyond its side information; minimizing it maximizes the eavesdropper’s equivocation about the source. In a storage reading the second party keeps the description compressed against what it already knows, and the same quantity is the least rate at which it can re-encode the description. Either way one description carries two costs, set by two different parties, and the encoder, which may observe more of the source than either party, decides how they trade off.

The single-letter region of jointly achievable rate, distortion, and equivocation is due to Villard and Piantanida [1, Theorem 3], and for vector Gaussian sources Ekrem and Uluks [2] obtained the minimum leakage when the rate is unconstrained, as an optimization over covariance matrices. The joint rate–leakage region was left open: they report that they could not solve the weighted problem that traces it [2, p. 8]. This paper solves that problem when the decoder has no side information. The encoder observes a jointly Gaussian vector T , the decoder must reproduce a linear part $Y = FT$ under a weighted quadratic distortion, and the second party observes $S = HT + U$ for an arbitrary matrix H . When H reaches the parts of T outside Y , the encoder holds information about the second party’s observation that the decoder does not need, and this is what makes the two costs disagree.

The results are structural as well as explicit. The coding theorem (Theorem 3) identifies the operational region with the single-letter one, and Gaussian descriptions, indexed by their error covariance, exhaust it (Proposition 2).

In that coordinate both costs are convex (Theorem 4), so every point of the rate–leakage frontier is the unique minimizer of a convex program and the minimum leakage is a determinant maximization under linear matrix inequalities. When the optimal description is one-dimensional, which an explicit certificate decides, the minimum leakage is

$$L(D) = \frac{1}{2} \log(1 + \Delta/\mu_{\max}),$$

with Δ the reduction in distortion the description must buy and $\mu_{\max}$ the largest generalized eigenvalue of a pencil formed from the source covariance, the decoder’s weight, the conditional covariance left after the second party’s observation, and the budget (Theorem 5). For a scalar described variable and a scalar context this is an explicit closed form (Corollary 6).

The eigenvector of the pencil is the direction the description reads, and it carries the paper’s main structural finding. The rate-optimal description reads the principal direction of the decoder’s weighted covariance at every budget, as do principal components, reverse water-filling [3], and the Gaussian information bottleneck [4]. The leakage-optimal description reads a direction that turns with the budget, unless that direction is a common generalized eigenvector of the decoder’s geometry and the second party’s (Theorem 8). No description optimized for a single party does this (Proposition 7). The same condition governs whether the two optimal descriptions coincide at all (Theorem 10): they differ whenever the second party’s observation touches the decoder’s principal direction. For independent components the leakage-optimal split of the distortion equalizes slopes that we compute in closed form, activates components in an order that is not the order of their variances, and can describe a disjoint set of components from the rate-optimal split (Section VI).

Related work

The conditional leakage studied here is the secrecy literature’s quantity. For a deterministic encoder, $H(T^n | M, S^n) = H(T^n | S^n) - H(M | S^n)$, so minimizing $H(M | S^n)/n$, which equals $I(T^n; M | S^n)/n$, is the same as maximizing the equivocation $H(T^n | M, S^n)/n$. Secure lossy source coding with side information at the eavesdropper was characterized by Villard and Piantanida [1]; with no decoder side information their Theorem 3 gives the discrete region of Theorem 15 below. Ekrem and Ulukus [2] treat vector Gaussian sources with side information at both receivers under a matrix distortion on the whole source, give an outer bound that is tight without a rate constraint, and leave the joint region open. Our distortion constrains a weighted part of the source and leaves the rest unpenalized, the decoder has no side information, and the eavesdropper’s observation map is arbitrary; in that setting the joint region is evaluated here in full. Related problems price the leakage of a description about side information [5], [6], [7] or the equivocation of the source in a cipher system [8], [9]. The leakage endpoint is Steinberg’s common-reconstruction functional [10] on the source as the encoder observes it, an augmentation noted for finite alphabets by Lapidoth, Malär, and Wigger [11]; a vector Gaussian common-reconstruction function with every component penalized was given by Lu and Xu [12]. Weighted quadratic distortion models a decoder that cares about some directions more than others, as in weighted rate–distortion [13], indirect source coding [14], and task-based quantization [15]. Parameterizing a Gaussian description by its error covariance is a standard device in vector Gaussian source coding [16], and the resulting endpoint program is a determinant maximization [17].

The paper is organized as follows. Section II sets up the model and the coding theorem, Section III proves the convex structure, Section IV the one-dimensional regime, Section V the results on directions and misalignment, and Section VI independent components. Section VII gives instances and Section VIII open problems. The appendices prove the coding theorem and a Gaussian exhaustion lemma and describe the numerical checks.

II. MODEL AND THE CODING THEOREM

Logarithms are base two unless written $\ln$. The encoder observes i.i.d. copies of $T \in \mathbb{R}^p$, jointly Gaussian and centered with $\Sigma_T \succ 0$. The described variable is $Y = FT \in \mathbb{R}^m$ for a matrix F of full row rank, and the decoder’s distortion is $\mathbb{E}[(Y - \hat{Y})^\top W(Y - \hat{Y})]$ for a weight $W \succ 0$. Write

$$Q := F^\top W F \succeq 0, \quad D_{\max} := \text{tr}(Q \Sigma_T).$$

The second party, which we call the holder, observes $S = HT + U \in \mathbb{R}^r$ with $U \sim \mathcal{N}(0, \Sigma_U)$, $\Sigma_U \succ 0$, independent of T and of all coding randomness. Write

$$J := H^\top \Sigma_U^{-1} H \succeq 0, \quad \Sigma_{T|S} = (\Sigma_T^{-1} + J)^{-1},$$

the information that S carries about T and the conditional covariance it leaves.

A code consists of an encoder $f_n : \mathbb{R}^{p \times n} \rightarrow \mathcal{M}_n$, a finite set, and a decoder $g_n : \mathcal{M}_n \rightarrow \mathbb{R}^{m \times n}$. With $M = f_n(T^n)$ and $\hat{\mathbf{Y}}^n = g_n(M)$, its rate, distortion, and leakage are

$$R_n = \frac{1}{n} \log |\mathcal{M}_n|, \quad D_n = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(Y_i - \hat{\mathbf{Y}}_i)^\top W (Y_i - \hat{\mathbf{Y}}_i)], \quad L_n = \frac{1}{n} H(M | S^n) = \frac{1}{n} I(T^n; M | S^n).$$

A pair (R, L) is achievable at distortion D when a sequence of codes has $\limsup R_n \leq R$, $\limsup L_n \leq L$, and $\limsup D_n \leq D$; the closure of the achievable set is $\mathcal{R}(D)$. Throughout $0 < D < D_{\max}$; at $D \geq D_{\max}$ the zero reproduction gives $R = L = 0$. Write $R_{\min}(D)$ and $L(D)$ for the least rate and the least leakage over $\mathcal{R}(D)$, and call a description rate-optimal or leakage-optimal when it attains one of them.

A test channel $p(\hat{\mathbf{y}} | t)$ is admissible when its distortion is at most D ; the Markov chain $\hat{\mathbf{Y}} - T - S$ holds because the description is formed without S . For a channel, let Σ_{e0} and Σ_e be the error covariances of the best linear estimates of T from $\hat{\mathbf{Y}}$ and from $(\hat{\mathbf{Y}}, S)$. Appendix B proves the following Gaussian exhaustion lemma.

Lemma 1. *Let $\hat{\mathbf{Y}} \in \mathbb{R}^m$ be centered with finite second moments, let A be its regression matrix on T and $N' = \hat{\mathbf{Y}} - AT$ its residual, and suppose $\text{Cov}(N', S) = 0$ and $\text{Cov } N' \succ 0$. Then $I(T; \hat{\mathbf{Y}}) \geq \frac{1}{2} \log(\det \Sigma_T / \det \Sigma_{e0})$ and $I(T; \hat{\mathbf{Y}} | S) \geq \frac{1}{2} \log(\det \Sigma_{T|S} / \det \Sigma_e)$, and the jointly Gaussian channel $AT + N$, $N \sim \mathcal{N}(0, \text{Cov } N')$ independent of (T, U) , has the same second moments and meets both bounds with equality.*

Proposition 2. *The union of the quadrants above $(I(T; \hat{\mathbf{Y}}), I(T; \hat{\mathbf{Y}} | S))$ over admissible channels equals the union, over $\Sigma_{e0} \in \mathcal{P}(D)$, of the quadrants above*

$$(r(\Sigma_{e0}), \ell(\Sigma_{e0})) = \left(\frac{1}{2} \log \frac{\det \Sigma_T}{\det \Sigma_{e0}}, \frac{1}{2} \log \frac{\det \Sigma_{T|S}}{\det \Sigma_e} \right), \quad \Sigma_e^{-1} = \Sigma_{e0}^{-1} + J, \quad (1)$$

where $\mathcal{P}(D) = \{\Sigma_{e0} : 0 \prec \Sigma_{e0} \preceq \Sigma_T, \text{tr}(Q\Sigma_{e0}) \leq D\}$.

Proof. Given an admissible channel, centering $\hat{\mathbf{Y}}$ changes neither coordinate and does not raise the distortion. If $\text{Cov } N'$ is singular, either a noiseless linear functional of T is revealed and both coordinates are infinite, or a nonzero combination of the components of $\hat{\mathbf{Y}}$ vanishes and may be dropped after an invertible linear map; so assume $\text{Cov } N' \succ 0$. The residual is uncorrelated with T by construction and with U by independence, hence with S , and Lemma 1 replaces the channel by $\hat{\mathbf{Y}} = AT + N$ with the same distortion and no larger coordinates. For it, $\Sigma_{e0}^{-1} = \Sigma_T^{-1} + A^\top \Sigma_N^{-1} A$, so $0 \prec \Sigma_{e0} \preceq \Sigma_T$; the pair $(\hat{\mathbf{Y}}, S)$ is a linear observation of T with independent noises, so $\Sigma_e^{-1} = \Sigma_{e0}^{-1} + J$; and the distortion is at least that of $\mathbb{E}[Y | \hat{\mathbf{Y}}]$, which is $\text{tr}(Q\Sigma_{e0})$. Conversely, given $\Sigma_{e0} \in \mathcal{P}(D)$, factor $\Sigma_{e0}^{-1} - \Sigma_T^{-1} = G^\top G$, let $Z = GT + N$ with $N \sim \mathcal{N}(0, I)$ independent of (T, U) , and reproduce $\hat{\mathbf{Y}} = \mathbb{E}[Y | Z]$. Then $\text{Cov}(T | Z) = \Sigma_{e0}$, the distortion is $\text{tr}(Q\Sigma_{e0}) \leq D$, $\hat{\mathbf{Y}} - T - S$ holds, and since $\hat{\mathbf{Y}}$ is a function of Z , data processing bounds its coordinates by those of Z , which are (1). $\square$

Theorem 3. *For every $0 < D < D_{\max}$, $\mathcal{R}(D)$ equals the union of Proposition 2.*

The proof is in Appendix A. Its discrete core is the region of Villard and Piantanida without decoder side information (Theorem 15); the Gaussian extension quantizes the side information, the source, and the reproduction and passes to the limit.

III. CONVEX STRUCTURE

Theorem 4. *On $\{0 \prec \Sigma_{e0} \preceq \Sigma_T\}$, the rate r and the leakage ℓ of (1) are strictly convex functions of Σ_{e0} , and the distortion $\text{tr}(Q\Sigma_{e0})$ is linear. Consequently:*

- (a) *for every $\alpha \in [0, 1]$ the program $\min\{\alpha r(\Sigma_{e0}) + (1 - \alpha)\ell(\Sigma_{e0}) : \Sigma_{e0} \in \mathcal{P}(D)\}$ has a unique minimizer, the distortion constraint is active there, and the minima over α trace the Pareto frontier of $\mathcal{R}(D)$;*
- (b) *in the coordinate Σ_e the minimum leakage is a determinant maximization,*

$$L(D) = \frac{1}{2} \log \det \Sigma_{T|S} - \frac{1}{2} \max\{\log \det \Sigma_e : 0 \prec \Sigma_e \preceq \Sigma_{T|S}, \text{tr}(Qh(\Sigma_e)) \leq D\}, \quad h(\Sigma_e) := (\Sigma_e^{-1} - J)^{-1}, \quad (2)$$

in which h is matrix-convex and the constraint is equivalent to a linear matrix inequality: with $J = B^\top B$, $\text{tr}(Qh(\Sigma_e)) \leq D$ if and only if some symmetric Z has $\text{tr}(QZ) \leq D$ and

$$\begin{bmatrix} Z - \Sigma_e & \Sigma_e B^\top \\ B \Sigma_e & I - B \Sigma_e B^\top \end{bmatrix} \succeq 0; \quad (3)$$

(c) a feasible Σ_e is the maximizer in (2) if and only if $\text{tr}(Qh(\Sigma_e)) = D$ and there are $\varpi \geq 0$ and $\Theta \succeq 0$ with

$$\frac{1}{2}\Sigma_e - \varpi \Sigma_{e0} Q \Sigma_{e0} = \Sigma_e \Theta \Sigma_e, \quad \Theta(\Sigma_{T|S} - \Sigma_e) = 0, \quad \Sigma_{e0} = h(\Sigma_e), \quad (4)$$

and then $\varpi > 0$.

Proof. Convexity. $r = \frac{1}{2} \log \det \Sigma_T - \frac{1}{2} \log \det \Sigma_{e0}$ is strictly convex. For the leakage, write $J = B^\top B$ and note that $\Sigma_e = (\Sigma_{e0}^{-1} + B^\top B)^{-1}$ is the error covariance of the best linear estimate of a vector with covariance Σ_{e0} from its noisy image under B with unit noise. For every matrix L , the estimator with gain L has error covariance $E_L(\Sigma_{e0}) = (I - LB)\Sigma_{e0}(I - LB)^\top + LL^\top \succeq \Sigma_e$, with equality at the optimal gain, and E_L is affine in Σ_{e0} . A pointwise minimum of affine maps in the Löwner order, attained at each point, is matrix-concave: at $\Sigma_t = (1-t)\Sigma_0 + t\Sigma_1$ with optimal gain L_t , $\Sigma_e(\Sigma_t) = E_{L_t}(\Sigma_t) = (1-t)E_{L_t}(\Sigma_0) + tE_{L_t}(\Sigma_1) \succeq (1-t)\Sigma_e(\Sigma_0) + t\Sigma_e(\Sigma_1)$. The map $\Sigma \mapsto -\log \det \Sigma$ is strictly convex and decreasing in the Löwner order, so $-\log \det \Sigma_e(\Sigma_t) \leq -\log \det((1-t)\Sigma_e(\Sigma_0) + t\Sigma_e(\Sigma_1)) \leq (1-t)(-\log \det \Sigma_e(\Sigma_0)) + t(-\log \det \Sigma_e(\Sigma_1))$, with strict inequality in the second step when $\Sigma_0 \neq \Sigma_1$, because $\Sigma_{e0} \mapsto \Sigma_e$ is injective. So ℓ is strictly convex.

(a) $\mathcal{P}(D)$ is convex, the objective is strictly convex and diverges as $\det \Sigma_{e0} \rightarrow 0$, and the feasible set is bounded, so the minimizer exists and is unique. Both r and ℓ decrease when Σ_{e0} increases in the Löwner order, so at a minimizer with slack distortion a move toward Σ_T would stay feasible and lower the objective, since Σ_T itself has distortion $D_{\max} > D$. The region is convex and upward closed, so its Pareto points are exactly the minima of the weighted programs.

(b) The map $\Sigma_{e0} \mapsto \Sigma_e$ is a bijection of $\{0 \prec \Sigma_{e0} \preceq \Sigma_T\}$ onto $\{0 \prec \Sigma_e \preceq \Sigma_{T|S}\}$, with inverse h : $\Sigma_{e0} \preceq \Sigma_T$ if and only if $\Sigma_e^{-1} = \Sigma_{e0}^{-1} + J \succeq \Sigma_T^{-1} + J = \Sigma_{T|S}^{-1}$. The matrix $B \Sigma_{T|S} B^\top$ has the same nonzero eigenvalues as $\Sigma_{T|S}^{1/2} J \Sigma_{T|S}^{1/2} \prec \Sigma_{T|S}^{1/2} (\Sigma_T^{-1} + J) \Sigma_{T|S}^{1/2} = I$, so on the feasible set $B \Sigma_e B^\top \prec I$, and the Woodbury identity gives $h(\Sigma_e) = \Sigma_e + \Sigma_e B^\top (I - B \Sigma_e B^\top)^{-1} B \Sigma_e$. By the Schur complement criterion [18, Sec. A.5.5], $h(\Sigma_e) \preceq Z$ if and only if (3) holds, and the block matrix there is affine in (Σ_e, Z) ; averaging two instances with $Z_i = h(\Sigma_e^{(i)})$ gives matrix convexity of h . Since $\text{tr}(QZ)$ is monotone in Z , the constraint $\text{tr}(Qh(\Sigma_e)) \leq D$ is equivalent to the existence of Z with $\text{tr}(QZ) \leq D$ and (3).

(c) The program (2) has a strictly feasible point ($\Sigma_{e0} = \epsilon \Sigma_T$ with $\epsilon D_{\max} < D$), so the Karush–Kuhn–Tucker conditions are necessary and sufficient. The differential of h is $dh = h \Sigma_e^{-1} d\Sigma_e \Sigma_e^{-1} h$, so the gradient of $\text{tr}(Qh)$ is $\Sigma_e^{-1} \Sigma_{e0} Q \Sigma_{e0} \Sigma_e^{-1}$, and stationarity of $-\frac{1}{2} \ln \det \Sigma_e + \varpi \text{tr}(Qh) + \text{tr}(\Theta(\Sigma_e - \Sigma_{T|S}))$, multiplied on both sides by Σ_e , is the first condition of (4), the factor $\ln 2$ being absorbed in ϖ . When the distortion constraint is active its complementary slackness holds for every $\varpi \geq 0$, and the maximizer has an active constraint by (a). If $\varpi = 0$, then $\Theta = \frac{1}{2} \Sigma_e^{-1} \succ 0$ and the second condition forces $\Sigma_e = \Sigma_{T|S}$, which is infeasible; so $\varpi > 0$. $\square$

IV. THE ONE-DIMENSIONAL REGIME

Write $\Delta = D_{\max} - D$ for the reduction in distortion the description must buy, and

$$A(\Delta) = \Sigma_T Q \Sigma_T - \Delta \Sigma_T.$$

A one-dimensional description is $Z = b^\top T + N$ with $b \neq 0$ and $N \sim \mathcal{N}(0, \sigma^2)$ independent, reproduced by $\mathbb{E}[Y | Z]$. Its read direction is b , and the direction in the space of the described variable along which it reduces the error is $F \Sigma_T b$.

Theorem 5. (a) A one-dimensional description meets the distortion constraint with finite leakage if and only if $\Delta < \lambda_{\max}(W^{1/2} \Sigma_Y W^{1/2})$, where $\Sigma_Y = F \Sigma_T F^\top$, and the least leakage among such descriptions is

$$L_{1d}(D) = \frac{1}{2} \log \left(1 + \frac{\Delta}{\mu_{\max}(\Delta)} \right), \quad (5)$$

where $\mu_{\max}(\Delta) > 0$ is the largest root of $\det(A(\Delta) - \mu \Sigma_{T|S}) = 0$. It is attained with b a corresponding generalized eigenvector and $\sigma^2 = b^\top \Sigma_T Q \Sigma_T b / \Delta - b^\top \Sigma_T b$, at which the distortion constraint is active.

(b) Suppose $\Delta < \lambda_{\max}(W^{1/2}\Sigma_Y W^{1/2})$, let Σ_{e0} and Σ_e be the error covariances of the description of (a), and let $\Gamma(\varpi) = \frac{1}{2}\Sigma_e - \varpi \Sigma_{e0} Q \Sigma_{e0}$. Then $L(D) = L_{1d}(D)$ if and only if there is $\varpi \geq 0$ with $\Gamma(\varpi) \succeq 0$ and $\Gamma(\varpi)b = 0$, and then $\varpi = \frac{1}{2}b^\top \Sigma_e b / (b^\top \Sigma_{e0} Q \Sigma_{e0} b)$.

(c) If the reproduction is scalar ($m = 1$), then $L(D) = L_{1d}(D)$ for every $0 < D < D_{\max}$.

Proof. (a) For the description Z , $\text{Cov}(T | Z) = \Sigma_T - \Sigma_T b b^\top \Sigma_T / (b^\top \Sigma_T b + \sigma^2)$, so the distortion of $\mathbb{E}[Y | Z]$ is $D_{\max} - b^\top \Sigma_T Q \Sigma_T b / (b^\top \Sigma_T b + \sigma^2)$, and the constraint reads $\sigma^2 \leq b^\top \Sigma_T Q \Sigma_T b / \Delta - b^\top \Sigma_T b$. The leakage $I(T; Z | S) = \frac{1}{2} \log(1 + b^\top \Sigma_{T|S} b / \sigma^2)$ decreases in σ^2 , so at the largest admissible σ^2 , where the constraint is active, it equals $\frac{1}{2} \log(1 + \Delta b^\top \Sigma_{T|S} b / b^\top A(\Delta) b)$, finite exactly when $b^\top A(\Delta) b > 0$. Minimizing over b is maximizing the Rayleigh quotient $b^\top A(\Delta) b / b^\top \Sigma_{T|S} b$ of a symmetric pencil with $\Sigma_{T|S} \succ 0$, whose maximum is $\mu_{\max}$. Some b has $b^\top A(\Delta) b > 0$ if and only if $\Sigma_T^{1/2} Q \Sigma_T^{1/2} \not\preceq \Delta I$, and the nonzero eigenvalues of that matrix are those of $W^{1/2} \Sigma_Y W^{1/2}$.

(b) The description of (a) has an active constraint and $\Sigma_e^{-1} = \Sigma_{T|S}^{-1} + b b^\top / \sigma^2$, so $\Sigma_{T|S} - \Sigma_e$ is a positive multiple of $\Sigma_{T|S} b b^\top \Sigma_{T|S}$ and $\Sigma_e^{-1} \Sigma_{T|S} b$ is a positive multiple of b . With $\Theta = \Sigma_e^{-1} \Gamma(\varpi) \Sigma_e^{-1}$, the conditions (4) at this point read $\Gamma(\varpi) \succeq 0$ and $\Gamma(\varpi)b = 0$, and Theorem 4(c) makes them necessary and sufficient. Taking $b^\top \Gamma(\varpi) b = 0$ gives ϖ .

(c) By Lemma 1 the optimum is attained by a jointly Gaussian channel $\hat{Y} = a^\top T + N$. If $\text{Cov}(Y, \hat{Y}) = 0$ its distortion is at least $D_{\max} > D$; otherwise $\mathbb{E}[Y | \hat{Y}]$ is a nonzero multiple of $\hat{Y}$, which leaves the leakage unchanged and does not raise the distortion, so the optimum is a one-dimensional description. $\square$

Corollary 6. Let $T = (Y, V)$ be scalar with unit variances and correlation ρ , $|\rho| < 1$, let $W = 1$, and let $S = V + U$ with $\text{Var } U = \tau^2 > 0$; write $s = 1 + \tau^2$. For $0 < D < 1$,

$$L(D) = \frac{1}{2} \log g^*, \quad g^* = \frac{(D + s - \rho^2) + \sqrt{(D + s - \rho^2)^2 - 4Ds(1 - \rho^2)}}{2Ds},$$

the larger root of $P(g) = Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2)$, attained by $\hat{Y} = aY + bV + N$ with $a = (g^* - 1)/g^*$, $b = (g^* - 1)\rho/(g^*(g^*s - 1))$, and the noise variance that makes the distortion equal to D .

Proof. By Theorem 5(c), $L(D) = \frac{1}{2} \log(1 + (1 - D)/\mu_{\max})$ with $\Delta = 1 - D$. Substituting $\mu = (1 - D)/(g - 1)$ into $\det(A(1 - D) - \mu \Sigma_{T|S})$, with $\Sigma_T = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$, $Q = e_1 e_1^\top$, and $J = e_2 e_2^\top / \tau^2$, gives $-(1 - D)(1 - \rho^2)P(g)/(s(g - 1)^2)$, a direct computation confirmed by computer algebra (Appendix C). Since $P(1) = (D - 1)\tau^2 < 0$ and $P(0) = 1 - \rho^2 > 0$, P has one root in $(0, 1)$, which corresponds to a negative μ , and one root $g^* > 1$, which corresponds to the unique positive root $\mu_{\max}$. The read direction is the corresponding generalized eigenvector; normalizing the reproduction gives the stated coefficients. $\square$

For a scalar described variable with vector context, Theorem 5(c) likewise gives $L(D)$ explicitly as the largest generalized eigenvalue of a pencil of size $1 + r$.

V. DIRECTIONS AND MISALIGNMENT

Proposition 7. With no holder ($J = 0$), the best one-dimensional description reads the top generalized eigenvector of the pencil $(\Sigma_T Q \Sigma_T, \Sigma_T)$ at every budget, and the rate-optimal description reverse water-fills on the eigenvalues of $W^{1/2} \Sigma_Y W^{1/2}$ along fixed directions.

Proof. With $J = 0$, $\Sigma_{T|S} = \Sigma_T$ and $A(\Delta)b = \mu \Sigma_T b$ reads $\Sigma_T Q \Sigma_T b = (\mu + \Delta) \Sigma_T b$, whose eigenvectors do not depend on Δ . The rate-optimal description is the weighted quadratic rate-distortion solution [13]: in the coordinates $W^{1/2} Y$ the distortion is squared error, and reverse water-filling describes fixed principal components [3]. $\square$

The Gaussian information bottleneck has the same property: its optimal projections are fixed eigenvectors, and its tradeoff parameter sets only their number and scale [4]. The leakage-optimal description behaves differently.

Theorem 8. Let $\mu_{\max}(\Delta)$ be simple and positive on an open interval $\mathcal{I} \subseteq (0, \lambda_{\max}(W^{1/2} \Sigma_Y W^{1/2}))$, with generalized eigenvector $b(\Delta)$. Then $b(\Delta)$ spans the same line for every $\Delta \in \mathcal{I}$ if and only if, for some $\Delta_0 \in \mathcal{I}$, $b_0 = b(\Delta_0)$ satisfies

$$Q \Sigma_T b_0 = \lambda b_0 \quad \text{and} \quad J \Sigma_T b_0 = \kappa b_0 \quad (6)$$

for scalars $\lambda > 0$ and $\kappa \geq 0$, that is, if and only if b_0 is a common generalized eigenvector of the pencils $(\Sigma_T Q \Sigma_T, \Sigma_T)$ and $(\Sigma_T J \Sigma_T, \Sigma_T)$. Otherwise $b(\Delta)$ is constant on no open subinterval of $\mathcal{I}$.

Proof. A simple eigenvalue of a symmetric pencil with positive definite second matrix, and its eigenvector, are analytic in the parameter [19]. Write $A(\Delta) = A_0 - \Delta \Sigma_T$.

Suppose b is constant on an open subinterval. Differentiating $(A_0 - \Delta \Sigma_T)b = \mu_{\max}(\Delta) \Sigma_{T|S} b$ gives $\Sigma_T b = c \Sigma_{T|S} b$ for a scalar c , and $c = b^\top \Sigma_T b / b^\top \Sigma_{T|S} b \geq 1$ because $\Sigma_{T|S} \preceq \Sigma_T$. Then $A_0 b = (\Delta + \mu_{\max}/c) \Sigma_T b$, that is, $Q \Sigma_T b = \lambda b$ with $\lambda = \Delta + \mu_{\max}/c > 0$; and multiplying $\Sigma_T b = c \Sigma_{T|S} b$ by $\Sigma_{T|S}^{-1} = \Sigma_T^{-1} + J$ gives $J \Sigma_T b = (c - 1)b$, so $\kappa = c - 1 \geq 0$. This proves the last sentence and the forward direction.

Conversely, if b_0 satisfies (6), then $\Sigma_{T|S}^{-1} \Sigma_T b_0 = (1 + \kappa)b_0$, so $\Sigma_T b_0 = (1 + \kappa) \Sigma_{T|S} b_0$ and $A(\Delta) b_0 = (\lambda - \Delta) \Sigma_T b_0 = (\lambda - \Delta)(1 + \kappa) \Sigma_{T|S} b_0$ for every Δ : b_0 is a generalized eigenvector with eigenvalue $(\lambda - \Delta)(1 + \kappa)$ throughout. Let $\mathcal{E} = \{\Delta \in \mathcal{I} : (\lambda - \Delta)(1 + \kappa) = \mu_{\max}(\Delta)\}$. It contains Δ_0 and is closed in $\mathcal{I}$ by continuity. It is open, because near a point of $\mathcal{E}$ the simple eigenvalue $\mu_{\max}$ is the only eigenvalue of the pencil close to it, and $(\lambda - \Delta)(1 + \kappa)$ is an eigenvalue close to it. Since $\mathcal{I}$ is connected, $\mathcal{E} = \mathcal{I}$, and by simplicity $b(\Delta)$ spans the line of b_0 throughout. $\square$

Corollary 9. Suppose the holder observes only the context: in coordinates $T = (Y, V)$, $H = [0 \ H_V]$. Then (6) holds if and only if $b_0 = F^\top \beta$ with $W \Sigma_Y \beta = \lambda \beta$ and $\text{Cov}(S, \beta^\top Y) = 0$.

Proof. $Q \Sigma_T b_0 = \lambda b_0$ with $\lambda > 0$ puts b_0 in the range of Q , which is the range of $F^\top$, so $b_0 = F^\top \beta$ and $F^\top W \Sigma_Y \beta = \lambda F^\top \beta$. The vector $J \Sigma_T b_0$ lies in the range of $H^\top$, which meets the range of $F^\top$ only at zero, so $J \Sigma_T b_0 = \kappa b_0$ forces $J \Sigma_T b_0 = 0$, that is, $\Sigma_U^{-1} H \Sigma_T F^\top \beta = 0$, which is $\text{Cov}(S, \beta^\top Y) = 0$. The converse is direct. $\square$

The same condition decides whether the two optimal descriptions coincide. Let b_R be the top generalized eigenvector of $(\Sigma_T Q \Sigma_T, \Sigma_T)$, the read direction of the best one-dimensional rate description.

Theorem 10. Suppose Δ lies in the one-dimensional regime of both the rate and the leakage problems. If $J \Sigma_T b_R$ is not parallel to b_R , the leakage-optimal description differs from the rate-optimal one, so each pays a strictly positive excess in the other cost. If $J \Sigma_T b_R$ is parallel to b_R , then b_R is a generalized eigenvector of the leakage pencil at every Δ , and the two descriptions coincide exactly when it is the top one.

Proof. $Q \Sigma_T b_R = \lambda_R b_R$ by definition of b_R . The vector b_R is a generalized eigenvector of $(A(\Delta), \Sigma_{T|S})$ if and only if $(\lambda_R - \Delta) \Sigma_T b_R \propto \Sigma_{T|S} b_R$, that is, $\Sigma_T b_R \propto \Sigma_{T|S} b_R$, which after multiplication by $\Sigma_{T|S}^{-1}$ is $J \Sigma_T b_R \propto b_R$. In the one-dimensional regime the two optimal descriptions are determined by their read directions and noise variances, and the minimizers of Theorem 4(a) are unique, so if the leakage-optimal read is not b_R the rate-optimal description is not leakage-optimal and conversely. $\square$

In the scalar case of Corollary 6, $b_R = e_1$ and $J \Sigma_T e_1 = (\rho/\tau^2) e_2$, which is parallel to e_1 only when $\rho = 0$. So the two descriptions differ exactly when the context is correlated with the described variable. The rate-optimal channel is the classical reverse channel, with leakage $\frac{1}{2} \log[((1 - D)(1 - \rho^2/s) + D)/D]$, strictly above $L(D)$, and the leakage-optimal channel of Corollary 6 has rate strictly above $\frac{1}{2} \log(1/D)$. The rate excess is unbounded toward the clean-context boundary $\tau^2 \rightarrow 0$ at $D = 1 - \rho^2$, where the leakage-optimal channel pins the reproduction to the context.

Corollary 11. There are sources for which the leakage-optimal one-dimensional read takes different directions at two budgets. For such a source no description problem for a single party with a fixed weighted quadratic distortion has the leakage-optimal descriptions as its optimal descriptions at both budgets.

Proof. Section VII gives an instance with reads at 47.48° and 41.15° at two budgets inside the one-dimensional regime. By Proposition 7 applied to any fixed weight, the best one-dimensional description for a single party reads the same direction at both budgets. $\square$

VI. INDEPENDENT COMPONENTS

Suppose $Y = (Y_1, \dots, Y_m)$, W is diagonal with entries w_j , and each Y_j has its own context V_j and holder observation $S_j = V_j + U_j$, the triples (Y_j, V_j, S_j) being independent, with $\text{Var } Y_j = \sigma_j^2$, $\text{Var } V_j = 1$, $\mathbb{E}[Y_j V_j] = \rho_j \sigma_j$,

$|\rho_j| < 1$, and $\text{Var } U_j = \tau_j^2 > 0$. Absorb w_j into σ_j^2 . Write ℓ_j for the scalar function $\delta \mapsto \frac{1}{2} \log g^*(\delta)$ of Corollary 6 at (ρ_j, τ_j^2) , extended by zero for $\delta \geq 1$, and $L_j(D_j) = \ell_j(D_j/\sigma_j^2)$.

Lemma 12. For $0 < \delta < 1$, with $g = g^*(\delta)$, $s = 1 + \tau^2$, $k = gs - 1$, and $\Delta_\delta = \sqrt{(\delta + s - \rho^2)^2 - 4\delta s(1 - \rho^2)} > 0$,

$$\ell'(\delta) = -\frac{k}{2 \ln 2 \Delta_\delta}, \quad \ell''(\delta) = \frac{(1 - \rho^2)k^3 + \rho^2 \tau^2 g^2 (\tau^4 g + \tau^2 (2g + 1) + g + 1)}{2 \ln 2 g \Delta_\delta^3} > 0,$$

so ℓ is strictly convex and decreasing, $\ell' \rightarrow -\infty$ as $\delta \rightarrow 0$, and $\ell' \rightarrow -\tau^2/(2 \ln 2(\rho^2 + \tau^2))$ as $\delta \rightarrow 1$.

Proof. The larger root is simple, with $\partial_g P = \Delta_\delta$ there, so g^* is smooth; differentiating $P(g^*(\delta)) = 0$ with $\partial_\delta P = gk$ gives $g' = -gk/\Delta_\delta$ and the first formula. Differentiating once more and eliminating δ through $\delta gk = (s - \rho^2)g - (1 - \rho^2)$ gives the second, an identity of polynomials confirmed by computer algebra (Appendix C); its numerator is positive. The limits follow from $g^* \rightarrow \infty$, $\Delta_\delta \rightarrow s - \rho^2$ as $\delta \rightarrow 0$ and $g^* \rightarrow 1$, $k \rightarrow \tau^2$, $\Delta_\delta \rightarrow \rho^2 + \tau^2$ as $\delta \rightarrow 1$. $\square$

Define the activation level $\kappa_j = -L'_j(\sigma_j^2) = \tau_j^2/(2 \ln 2 \sigma_j^2(\rho_j^2 + \tau_j^2))$; for the rate it would be $1/(2 \ln 2 \sigma_j^2)$.

Theorem 13. (a) $\mathcal{R}(D)$ is the union, over $D_j \geq 0$ with $\sum_j D_j \leq D$, of the Minkowski sums of the scalar regions of the components, attained by product channels. (b) $L(D) = \min \sum_j L_j(D_j)$ over such allocations, with a unique minimizer $D_j = D_j(\theta^*)$, where $D_j(\theta) = \sigma_j^2$ for $\theta \leq \kappa_j$ and otherwise solves $-L'_j(D_j) = \theta$, and θ^* is the unique level with $\sum_j D_j(\theta^*) = D$. (c) As D decreases, the leakage-optimal description begins to describe component j in decreasing order of $\sigma_j^2(1 + \rho_j^2/\tau_j^2)$, while the rate-optimal description proceeds in decreasing order of σ_j^2 .

Proof. (a) By Theorem 3 it suffices to identify the single-letter union. For an admissible channel with $D_j = \mathbb{E}(Y_j - \hat{Y}_j)^2$, each induced channel $p(\hat{y}_j | t_j)$ satisfies $\hat{Y}_j - T_j - S_j$, and since the pairs (T_j, S_j) are independent the chain rule gives $I(T; \hat{\mathbf{Y}}) \geq \sum_j I(T_j; \hat{Y}_j)$ and $I(T; \hat{\mathbf{Y}} | S) = \sum_j [I(T_j; \hat{\mathbf{Y}}, S, T^{j-1}) - I(T_j; S_j)] \geq \sum_j I(T_j; \hat{Y}_j | S_j)$. A product of component channels attains the sum with equality. (b) Each L_j is convex and nonincreasing, strictly convex on $(0, \sigma_j^2)$ by Lemma 12, with left derivative $-\kappa_j$ at σ_j^2 . An allocation is optimal if and only if some $\theta \geq 0$ lies in $-\partial L_j(D_j)$ for every j with $\theta(\sum_j D_j - D) = 0$. Since $D < \sum_j \sigma_j^2$, $\theta > 0$; no D_j is 0 because $-L'_j \rightarrow \infty$ there; and $\sum_j D_j(\theta)$ decreases continuously and strictly from $\sum_j \sigma_j^2$ at $\min_j \kappa_j$ to 0, so θ^* is unique. (c) $D_j(\theta) < \sigma_j^2$ exactly when $\theta > \kappa_j$, and θ^* increases as D decreases. $\square$

Corollary 14. If $m = 2$, $\sigma_1^2 > \sigma_2^2$, and $\sigma_1^2(1 + \rho_1^2/\tau_1^2) < \sigma_2^2(1 + \rho_2^2/\tau_2^2)$, then for every $D \in [\max\{2\sigma_2^2, \sigma_1^2 + D_2(\kappa_1)\}, \sigma_1^2 + \sigma_2^2]$ the rate-optimal description describes component 1 alone and the leakage-optimal description describes component 2 alone.

Proof. Reverse water-filling describes component 1 alone exactly for $2\sigma_2^2 \leq D < \sigma_1^2 + \sigma_2^2$; Theorem 13(b) describes component 2 alone exactly when $\kappa_2 < \theta^* \leq \kappa_1$, that is, for $\sigma_1^2 + D_2(\kappa_1) \leq D < \sigma_1^2 + \sigma_2^2$. $\square$

VII. INSTANCES

Scalar. At $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$ the rate-optimal channel pays 0.0349 bits of excess leakage and the leakage-optimal channel 0.0400 bits of excess rate; over the box $\rho^2 \in [0.05, 0.95]$, $\tau^2 \in [\frac{1}{4}, 4]$, $D \in [0.05, 0.95]$ the rate excess of the leakage-optimal channel stays below 0.114 bits and the leakage excess of the rate-optimal channel below 0.077 bits, and at $(\rho^2, \tau^2, D) = (0.5, 10^{-3}, 0.5)$ the rate excess reaches 1.537 bits. Two sources with the same law of (Y, S) up to scaling of S , $(\rho^2, \tau^2) = (\frac{3}{5}, \frac{1}{5})$ and $(\frac{3}{4}, \frac{1}{2})$, have $L = 1.1880$ and 1.2105 bits at $D = 0.1$: the leakage is not determined by the law of (Y, S) , because the encoder observes the context.

Coupled. Take $\Sigma_Y = \text{diag}(1, 0.4)$, $W = I$, a scalar context with $\text{Var } V = 1$ and $\text{Cov}(Y, V) = 0.6u$ for the unit vector u at 50° , and $S = V + U$ with $\text{Var } U = 0.05$. The certificate of Theorem 5(b) holds for $D \geq 1.08530$, out of $D_{\max} = 1.4$. The rate-optimal description reduces the error along the first axis for every $D \geq 0.8$. The leakage-optimal description reduces it along 47.48° , 46.76° , 45.07° , and 41.15° at $D = 1.38, 1.3, 1.2$, and 1.1 . Four sources with the same law of (Y, S) up to scaling, $(|\text{Cov}(Y, V)|^2, \tau^2) = (0.33, 0.1)$, $(0.36, 0.2)$, $(0.39, 0.3)$, $(0.42, 0.4)$, read along 42.83° , 37.61° , 33.75° , and 30.84° at $D = 1.3$: the direction, too, is not determined by the law of (Y, S) .

Independent components. With $(\sigma_1^2, \rho_1^2, \tau_1^2) = (1, 0, 1)$ and $(\sigma_2^2, \rho_2^2, \tau_2^2) = (0.6, 0.9, 0.05)$, $\kappa_1 = 0.7213$, $\kappa_2 = 0.0633$, and the two descriptions describe disjoint components for every $D \in [1.2002, 1.6)$. At $D = 1.4$ the

rate-optimal description pays $R = L = 0.1610$ bits on component 1 and the leakage-optimal description pays $L = 0.0197$ and $R = 0.3272$ bits on component 2.

VIII. DISCUSSION

A description held by two parties has a cost for each, and the encoder's observations decide how the costs trade off. For Gaussian sources without decoder side information the region is now explicit in structure: a convex program in the error covariance, a closed form in a one-dimensional regime, and directions that turn with the budget exactly when the two parties' geometries are not aligned. Several problems remain open. With side information at the decoder the weighted problem of Ekrem and Ulukus remains unsolved, because the Gaussian exhaustion lemma no longer handles both costs at once once the decoder bins against its own observation. A closed form below the one-dimensional regime, and for its boundary as a function of the source, is not known. Stationary sources and finite blocklengths are not treated.

APPENDIX A THE CODING THEOREM

A. Finite alphabets

Let (X, S) be a discrete memoryless pair with finite alphabets and let d be a distortion with values in $[0, d_{\max}]$; codes, rate, distortion, and leakage $H(M | S^n)/n$ are as in Section II.

Theorem 15 ([1, Theorem 3], without decoder side information). *The closure of the achievable set at distortion D is $\bigcup_{p(\hat{x}|x): \mathbb{E}d(X, \hat{X}) \leq D} \{(R, L) : R \geq I(X; \hat{X}), L \geq I(X; \hat{X} | S)\}$, and the union is convex and closed.*

Proof outline. Villard and Piantanida prove the region in the equivocation form $\Delta \leq H(X | V) + 0 - I(X; S | U)$, $U - V - X - S$, with no decoder side information; taking $U = V$ is optimal, because $I(X; S | U) = I(V; S | U) + I(X; S | V) \geq I(X; S | V)$, and $\Delta = H(X | S) - L$ for a deterministic encoder, which gives the displayed form with V in place of $\hat{X}$, and a reproduction that is a function of V only lowers both coordinates. For the Gaussian extension we use the following finite-blocklength form of the achievability, obtained by covering at rate $I(X; \hat{X}) + 2\epsilon$, a random binning of the codeword index into $2^{n(I(X; \hat{X}|S) + 3\epsilon)}$ bins used only as a proof device, Fano's inequality for a hypothetical decoder holding the bin index and S^n , and selection of one deterministic code: for every channel with finite reproduction alphabet and every $\epsilon > 0$ there are $\epsilon_n \rightarrow 0$ and, for all large n , a deterministic code with $|\mathcal{M}_n| \leq 2^{n(I(X; \hat{X}) + 2\epsilon)}$, distortion at most $\mathbb{E}d(X, \hat{X}) + \epsilon + 2\epsilon_n d_{\max}$, and $H(M | S^n) \leq n(I(X; \hat{X} | S) + 3\epsilon) + 1 + 2\epsilon_n n(I(X; \hat{X}) + 2\epsilon)$. Convexity of both coordinates in the channel, for the conditional one by the decomposition $I(X; \hat{X} | S) = \sum_s p(s) I_{p(x|s)}(q)$, makes the union convex without time sharing. $\square$

B. The Gaussian source

Proof of Theorem 3. Converse. For any code, $\log |\mathcal{M}_n| \geq I(T^n; \hat{\mathbf{Y}}^n) \geq \sum_i I(T_i; \hat{\mathbf{Y}}_i)$ and $H(M | S^n) \geq I(T^n; \hat{\mathbf{Y}}^n | S^n) = \sum_i [I(T_i; \hat{\mathbf{Y}}^n, T^{i-1}, S^n) - I(T_i; S_i)] \geq \sum_i I(T_i; \hat{\mathbf{Y}}_i | S_i)$, by the chain rule and the independence of the pairs (T_i, S_i) ; every term is finite, being at most $I(T_i; S_i) + \log |\mathcal{M}_n|$ with $I(T_i; S_i) = \frac{1}{2} \log \det(H\Sigma_T H^\top + \Sigma_U) / \det \Sigma_U$. The mixture of the per-letter channels satisfies the Markov chain, has distortion D_n , and by convexity of both coordinates in the channel (for general alphabets the conditional coordinate is an integral over s of relative entropies affine in the channel [23], [22]) lies below (R_n, L_n) , so (R_n, L_n) is in the union at D_n .

Closedness. Let $(R_j, L_j) \rightarrow (R, L)$ lie in the union at levels D_j with $\limsup D_j \leq D$, with $\Sigma_j \in \mathcal{P}(D_j)$ below them by Proposition 2. The Σ_j lie in the compact set $0 \preceq \Sigma \preceq \Sigma_T$ and $\det \Sigma_j \geq \det \Sigma_T 2^{-2R_j}$ is bounded away from zero, so a subsequence converges to some $\Sigma^* \succ 0$ in $\mathcal{P}(D)$; r and ℓ are continuous on positive definite matrices, so (R, L) lies above $(r(\Sigma^*), \ell(\Sigma^*))$.

Achievability. Fix an admissible channel with $I(T; \hat{\mathbf{Y}}) < \infty$; then $\mathbb{E}\|\hat{\mathbf{Y}}\|^2 < \infty$. Let q_k be the saturating dyadic quantizer of level k applied coordinatewise: the bounded cells $[j2^{-k}, (j+1)2^{-k})$, $-4^k \leq j < 4^k$, map to their left endpoints and the two tail cells to $\pm 2^k$; the partitions are nested and generate the Borel σ -field.

Step A. The σ -fields of $S_\eta := q_\eta(S)$ increase to $\sigma(S)$, so $I(\hat{\mathbf{Y}}; S_\eta) \uparrow I(\hat{\mathbf{Y}}; S)$ [20, Ch. 2]; fix η with $I(\hat{\mathbf{Y}}; S_\eta) \geq I(\hat{\mathbf{Y}}; S) - \epsilon$. Since $\hat{\mathbf{Y}} - T - S_\eta$, $I(T; \hat{\mathbf{Y}} | S_\eta) = I(T; \hat{\mathbf{Y}}) - I(\hat{\mathbf{Y}}; S_\eta) \leq I(T; \hat{\mathbf{Y}} | S) + \epsilon$.

Step B. With $T_k = q_k(T)$ and $\hat{\mathbf{Y}}_k = q_k(\hat{\mathbf{Y}})$, the surrogate source (T_k, S_η) is discrete memoryless, the surrogate channel is $p(\hat{\mathbf{y}} | t_k) = \Pr(\hat{\mathbf{Y}}_k = \hat{\mathbf{y}} | T_k = t_k)$, and the surrogate distortion $d_k(t_k, \hat{\mathbf{y}}) = \mathbb{E}[(Y - \hat{\mathbf{y}})^\top W(Y - \hat{\mathbf{y}}) | T_k = t_k]$ is bounded for each k , every cell having positive probability since $\Sigma_T \succ 0$. The finite-blocklength achievability of Theorem 15 gives a code of rate at most $I(T_k; \hat{\mathbf{Y}}_k) + 2\epsilon \leq I(T; \hat{\mathbf{Y}}) + 2\epsilon$. Its true distortion equals its surrogate distortion, because given T_k^n the law of Y_i depends on $T_{k,i}$ alone. The surrogate distortion of the channel converges to $\mathbb{E}[(Y - \hat{\mathbf{Y}})^\top W(Y - \hat{\mathbf{Y}})] \leq D$, because $\mathbb{E}[Y | T_k] \rightarrow Y$ in L^2 by martingale convergence and $\hat{\mathbf{Y}}_k \rightarrow \hat{\mathbf{Y}}$ in L^2 . The surrogate law of $(\hat{\mathbf{Y}}_k, S_\eta)$ converges weakly to the true law of $(\hat{\mathbf{Y}}, S_\eta)$, by the bounded-convergence and martingale argument applied to test functions bounded on $\mathbb{R}^m \times q_\eta(\mathbb{R}^r)$ and continuous in the first argument, and the Markov chain $\hat{\mathbf{Y}} - T - S$. Mutual information is lower semicontinuous under weak convergence [21], so $I_k(T_k; \hat{\mathbf{Y}}_k | S_\eta) = I(T_k; \hat{\mathbf{Y}}_k) - I_k(\hat{\mathbf{Y}}_k; S_\eta) \leq I(T; \hat{\mathbf{Y}} | S) + 2\epsilon$ for large k , and the code has $H(M | S_\eta^n) \leq n(I(T; \hat{\mathbf{Y}} | S) + 5\epsilon + o(1))$.

Step C. S_η^n is a function of S^n , so $H(M | S^n) \leq H(M | S_\eta^n)$. A diagonal sequence $\epsilon \downarrow 0$ achieves the corner $(I(T; \hat{\mathbf{Y}}), I(T; \hat{\mathbf{Y}} | S))$ at distortion D , and with the converse, closedness, and Proposition 2 the region is as stated. $\square$

APPENDIX B

THE GAUSSIAN EXHAUSTION LEMMA

Proof of Lemma 1. The divergence bound. Conditional mutual information is the expected divergence of the conditional law of T given $(\hat{\mathbf{Y}}, S)$ from the conditional law given S [22], [20]. Fix a realization, let P be the conditional law with mean μ and covariance C , $P_G = \mathcal{N}(\mu, C)$, and $Q_S = \mathcal{N}(\mu_S, \Sigma_{T|S})$. If P has no density the divergence is infinite. Otherwise $\log(dP_G/dQ_S)$ is quadratic in t and P, P_G share two moments, so $D(P||Q_S) = D(P||P_G) + D(P_G||Q_S) \geq D(P_G||Q_S)$. Averaging $D(P_G||Q_S)$, the trace term averages to p by the law of total covariance, and by concavity of $\ln \det$ and Jensen, $\mathbb{E}[-\ln \det C] \geq -\ln \det \mathbb{E}C \geq -\ln \det \Sigma_e$, since the conditional mean has error covariance no larger than any linear estimator. Hence $I(T; \hat{\mathbf{Y}} | S) \geq \frac{1}{2} \log(\det \Sigma_{T|S} / \det \Sigma_e)$, and the same argument without S gives $I(T; \hat{\mathbf{Y}}) \geq \frac{1}{2} \log(\det \Sigma_T / \det \Sigma_{e0})$.

The determinant identity. With $\hat{\mathbf{Y}} = AT + N'$, $\text{Cov}(N', T) = 0$, and $\text{Cov}(N', S) = 0$, the change of variables $(T, S, \hat{\mathbf{Y}}) \mapsto (T, S, N')$ is block triangular with unit diagonal, so $\det \text{Cov}(T, S, \hat{\mathbf{Y}}) = \det \Sigma_S \det \Sigma_{T|S} \det \Sigma_N$; the Schur factorization in the other order gives $\det \text{Cov}(\hat{\mathbf{Y}}, S) \det \Sigma_e$, and $\text{Cov}(\hat{\mathbf{Y}}) - \text{Cov}(\hat{\mathbf{Y}}, S) \Sigma_S^{-1} \text{Cov}(S, \hat{\mathbf{Y}}) = A \Sigma_{T|S} A^\top + \Sigma_N \succ 0$. So $\det \Sigma_e = \det \Sigma_{T|S} \det \Sigma_N / \det(A \Sigma_{T|S} A^\top + \Sigma_N)$, finite and positive.

Equality. For the jointly Gaussian channel $AT + N$ the conditional law of T given $(\hat{\mathbf{Y}}, S)$ is Gaussian with the linear estimate as mean and Σ_e as covariance, so every inequality is an equality; its second moments match the given channel's. $\square$

APPENDIX C

NUMERICAL CHECKS

The identities and instances were checked independently of the derivations by scripts run on a remote workstation. `verify_general.py` checks, for random sources with general weights and observation maps, that the pencil value of Theorem 5 equals the brute-force optimum over Gaussian descriptions whenever that optimum is one-dimensional, that both costs are convex in the error covariance (16000 midpoint tests), that the weighted minimizers are unique, the two directions of Theorem 8, and the criterion of Theorem 10. `verify_coupled.py`, `verify_parallel.py`, and `verify_pencil.py` check the scalar identity of Corollary 6 symbolically, the determinant program against brute force, the certificate, Lemma 12, the allocation, and the numbers of Section VII. No proof depends on them.

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